3.9.3 Cosmology (A-level only)

3.9.3.1 Doppler effect (A-level only)

Content
$\frac{\Delta f}{f} = \frac{v}{c}$ and $z = \frac{\Delta \lambda}{\lambda} = -\frac{v}{c}$ for $v \ll c$ applied to optical and radio frequencies. Calculations on binary stars viewed in the plane of orbit. Galaxies and quasars.

Red shift $v = Hd$

Simple interpretation as expansion of universe; estimation of age of universe, assuming H is constant.

Qualitative treatment of Big Bang theory including evidence from cosmological microwave background radiation, and relative abundance of hydrogen and helium.

3.9.3.3 Quasars (A-level only)

Content
Quasars as the most distant measurable objects.
Discovery of quasars as bright radio sources.
Quasars show large optical red shifts; estimation involving distance and power output.
Formation of quasars from active supermassive black holes.

3.9.3.4 Detection of exoplanets (A-level only)

Content
Difficulties in the direct detection of exoplanets.
Detection techniques will be limited to variation in Doppler shift (radial velocity method) and the transit method.
Typical light curve.